

WEB PROGRAMMING LAB – Viva oriented questions

1. What is the Web?

Answer:

The Web (World Wide Web) is a system of interlinked hypertext documents accessed over the Internet using web browsers and web servers through the HTTP/HTTPS protocol.

2. Explain the client/server concept.

Answer:

In the client/server model, the **client** (browser) sends requests for services or resources, and the **server** processes the request and sends back a response. This separation improves scalability and efficiency.

3. What is a web client?

Answer:

A web client is an application (such as Chrome, Firefox, or Edge) that sends HTTP requests to web servers and displays the received web content to users.

4. What is a web server?

Answer:

A web server is a system that stores, processes, and delivers web pages to clients using HTTP/HTTPS (e.g., Apache, Nginx, IIS).

5. What are the main components of a web application?

Answer:

The main components are:

- Client (Browser)
- Web Server
- Application Server
- Database
- Network/Internet

6. What is static web content?

Answer:

Static web content consists of fixed files (HTML, CSS, images) that do not change unless manually updated. The server sends the same content to all users.

7. What is dynamic web content?

Answer:

Dynamic web content is generated at runtime based on user input, database data, or application logic using server-side technologies.

8. Give examples of dynamic web technologies.

Answer:

Examples include:

- PHP
- JSP

- ASP.NET
- Node.js
- Python (Django, Flask)

9. What is HTTP?

Answer:

HTTP (Hypertext Transfer Protocol) is a stateless application-layer protocol used for communication between web clients and servers.

10. What is an HTTP request?

Answer:

An HTTP request is sent by a client to a server to request a resource. It contains:

- Request method (GET, POST, etc.)
- URL
- Headers
- Optional body

11. What is an HTTP response?

Answer:

An HTTP response is sent by the server to the client and includes:

- Status code
- Headers
- Response body (requested data)

12. List common HTTP request methods.

Answer:

Common HTTP methods include:

- GET
- POST
- PUT
- DELETE
- PATCH
- HEAD

13. What are HTTP status codes?

Answer:

HTTP status codes indicate the result of a request, such as:

- 200 – OK
- 404 – Not Found
- 500 – Internal Server Error

14. Explain the HTTP request–response cycle.

Answer:

The client sends an HTTP request → the server processes it → the server sends an HTTP response → the client displays the content.

15. What is an application server?

Answer:

An application server hosts business logic, processes dynamic requests, connects to databases, and generates dynamic web pages.

16. Difference between web server and application server.

Answer:

A web server handles HTTP requests and static content, while an application server executes application logic and generates dynamic content.

17. How are dynamic web pages generated?

Answer:

Dynamic web pages are generated using server-side scripts that process user input, interact with databases, and create HTML at runtime.

18. What is web security?

Answer:

Web security refers to protecting web applications and data from threats like hacking, data theft, and unauthorized access.

19. What are common web security threats?

Answer:

Common threats include:

- SQL Injection
- Cross-Site Scripting (XSS)
- Cross-Site Request Forgery (CSRF)
- Malware

20. List common web security measures.

Answer:

Security measures include:

- HTTPS/SSL
- Authentication and authorization
- Input validation
- Firewalls
- Secure cookies

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1. What is HTML?

Answer:

HTML (HyperText Markup Language) is a markup language used to create and structure web pages.

2. What are HTML tags?

Answer:

HTML tags are keywords enclosed in angle brackets < > used to define elements in a web page.

3. How do you format text in HTML?

Answer:

Text formatting is done using tags like , <i>, <u>, , , and <mark>.

4. How do you change font size and style in HTML?

Answer:

Using CSS styles or the tag (deprecated), for example using style="font-size:16px;".

5. How do you write comments in HTML?

Answer:

Comments are written using <!-- Comment -->.

6. What is an anchor tag?

Answer:

The anchor tag <a> is used to create hyperlinks in HTML.

7. How do you create a hyperlink?

Answer:

Using the anchor tag with href attribute:

```
<a href="url">Link</a>
```

8. What is the use of the target attribute in hyperlinks?

Answer:

It specifies where the linked document opens, e.g., _blank, _self.

9. How do you set a background color in HTML?

Answer:

By using CSS:

```
<body style="background-color:blue;">
```

10. How do you add a background image in HTML?

Answer:

Using CSS:

```
<body style="background-image:url('image.jpg');">
```

11. What is the `<img>` tag?

Answer:

The `<img>` tag is used to insert images into a web page.

12. What attributes are used with the `<img>` tag?

Answer:

Common attributes are `src`, `alt`, `width`, and `height`.

13. What are HTML lists?

Answer:

Lists are used to group related items and are of three types: ordered, unordered, and description lists.

14. Difference between ordered and unordered lists.

Answer:

Ordered lists use numbers (`<ol>`), while unordered lists use bullets (`<ul>`).

15. What is an HTML table?

Answer:

An HTML table is used to display data in rows and columns using `<table>`, `<tr>`, `<td>`, and `<th>` tags.

16. What is the use of `<th>` tag?

Answer:

`<th>` defines a header cell in a table and displays text in bold and centered by default.

17. What are frames in HTML?

Answer:

Frames divide the browser window into multiple sections, each displaying a separate HTML document.

18. Which tags are used to create frames?

Answer:

`<frameset>` and `<frame>` tags (now deprecated in HTML5).

19. What is an HTML form?

Answer:

An HTML form is used to collect user input and send it to a server.

20. What are common form elements?

Answer:

Common form elements include:

- `<input>`
- `<textarea>`
- `<select>`
- `<button>`

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1. What is CSS?

Answer:

CSS (Cascading Style Sheets) is used to style and control the layout and appearance of HTML elements.

2. Why do we need CSS?

Answer:

CSS separates content from presentation, improves design consistency, reduces code repetition, and makes web pages easier to maintain.

3. What are the advantages of using CSS?

Answer:

Faster page loading, better control over layout, reusability of styles, and improved accessibility.

4. What is the basic syntax of CSS?

Answer:

```
selector { property: value; }
```

5. What is a CSS selector?

Answer:

A selector is used to select the HTML element(s) that need to be styled.

6. What is inline CSS?

Answer:

Inline CSS is applied directly inside an HTML tag using the `style` attribute.

7. Give an example of inline CSS.

Answer:

```
<p style="color:red;">Text</p>
```

8. What is embedded (internal) CSS?

Answer:

Embedded CSS is written inside the `<style>` tag within the `<head>` section of an HTML document.

9. What is external CSS?

Answer:

External CSS is written in a separate `.css` file and linked to HTML using the `<link>` tag.

10. Why is external CSS preferred?

Answer:

It allows reusability, easy maintenance, and consistent styling across multiple web pages.

11. How do you link an external CSS file?

Answer:

Using:

```
<link rel="stylesheet" href="style.css">
```

12. How do you set a background color using CSS?

Answer:

Using the `background-color` property.

13. How do you add a background image using CSS?

Answer:

Using `background-image: url("image.jpg");`

14. Which CSS properties are used to manipulate text?

Answer:

`color`, `font-size`, `font-family`, `text-align`, `text-decoration`, `text-transform`.

15. What is margin in CSS?

Answer:

Margin controls the space outside an element, between the element and other elements.

16. What is padding in CSS?

Answer:

Padding controls the space inside an element, between the content and the border.

17. Difference between margin and padding.

Answer:

Margin affects outer spacing, while padding affects inner spacing of an element.

18. What is the CSS box model?

Answer:

It consists of content, padding, border, and margin.

19. What is CSS positioning?

Answer:

CSS positioning controls how elements are placed on a web page.

20. What are the types of CSS positioning?

Answer:

`static`, `relative`, `absolute`, `fixed`, and `sticky`.

1. What is JavaScript?

Answer:

JavaScript is a client-side scripting language used to create dynamic and interactive web pages.

2. What are the core features of JavaScript?

Answer:

Lightweight, interpreted, object-based, event-driven, platform-independent, and supports client-side validation.

3. Where is JavaScript code written in an HTML page?

Answer:

Inside the `<script>` tag, either in the `<head>`, `<body>`, or in an external `.js` file.

4. What are JavaScript data types?

Answer:

Number, String, Boolean, Undefined, Null, Object, and Symbol.

5. How do you declare variables in JavaScript?

Answer:

Using `var`, `let`, or `const`.

6. Difference between `var`, `let`, and `const`.

Answer:

`var` has function scope, `let` has block scope, and `const` is block-scoped and cannot be reassigned.

7. What are operators in JavaScript?

Answer:

Operators are symbols used to perform operations on variables and values.

8. Name types of operators in JavaScript.

Answer:

Arithmetic, Relational, Logical, Assignment, and Conditional operators.

9. What is an expression?

Answer:

An expression is a combination of values, variables, and operators that produces a result.

10. What is a statement in JavaScript?

Answer:

A statement is an instruction that performs an action, such as assignment or conditional execution.

11. What is a function in JavaScript?

Answer:

A function is a reusable block of code designed to perform a specific task.

12. How are functions defined in JavaScript?

Answer:

Using the `function` keyword or arrow functions.

13. What is an object in JavaScript?

Answer:

An object is a collection of key-value pairs used to store related data and methods.

14. What is an array?

Answer:

An array is a special object used to store multiple values in a single variable.

15. What is the String object?

Answer:

The String object is used to manipulate text, such as finding length, converting case, or extracting characters.

16. What is the Date object?

Answer:

The Date object is used to work with dates and times in JavaScript.

17. What is the Math object?

Answer:

The Math object provides mathematical functions like `sqrt()`, `round()`, `random()`, and `max()`.

18. What is the Document Object Model (DOM)?

Answer:

DOM is a programming interface that represents the HTML document as a tree structure and allows JavaScript to manipulate web page content.

19. What is event handling in JavaScript?

Answer:

Event handling is the process of responding to user actions like clicks, key presses, or mouse movements.

20. What is form validation?

Answer:

Form validation ensures that user input meets required conditions before submitting the form.

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1. What is a relational database?

Answer:

A relational database stores data in the form of tables (relations) consisting of rows and columns.

2. What is a table in a relational database?

Answer:

A table is a collection of related data organized in rows and columns.

3. What is an attribute?

Answer:

An attribute is a column in a table that represents a property of an entity.

4. What is a tuple?

Answer:

A tuple is a single row in a table that represents one record.

5. What is a primary key?

Answer:

A primary key is a field or combination of fields that uniquely identifies each record in a table.

6. What are the characteristics of a primary key?

Answer:

It must be unique, cannot be NULL, and should be stable.

7. What is a foreign key?

Answer:

A foreign key is a field in one table that refers to the primary key of another table.

8. Why are foreign keys used?

Answer:

They maintain referential integrity between related tables.

9. What is an index?

Answer:

An index is a database object used to speed up data retrieval operations.

10. What are DDL commands?

Answer:

DDL (Data Definition Language) commands are used to define and modify database structures.

11. What is the CREATE command?

Answer:

CREATE is used to create database objects like tables and indexes.

12. What is the ALTER command?

Answer:

ALTER is used to modify the structure of an existing table.

13. What is the DROP command?

Answer:

DROP is used to permanently delete database objects from the database.

14. What is the TRUNCATE command?

Answer:

TRUNCATE removes all records from a table quickly without logging individual row deletions.

15. Difference between DROP and TRUNCATE.

Answer:

DROP removes the table structure and data, while TRUNCATE removes only data and keeps the structure.

16. What are DML commands?

Answer:

DML (Data Manipulation Language) commands are used to manage data stored in tables.

17. What is the SELECT command?

Answer:

SELECT retrieves data from one or more tables.

18. What is the INSERT command?

Answer:

INSERT adds new records into a table.

19. What is the UPDATE command?

Answer:

UPDATE modifies existing records in a table.

20. What is the DELETE command?

Answer:

DELETE removes specific records from a table based on a condition.

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1. What is PHP?

Answer:

PHP (Hypertext Preprocessor) is a server-side scripting language used to create dynamic web pages.

2. Why is PHP called a server-side language?

Answer:

Because PHP code is executed on the server and the output (HTML) is sent to the client browser.

3. What is XAMPP?**Answer:**

XAMPP is a local server package that includes Apache, MySQL, PHP, and Perl for web development.

4. What are the components of XAMPP?**Answer:**

Apache (web server), MySQL/MariaDB (database), PHP, and Perl.

5. How do you write PHP code?**Answer:**

PHP code is written inside `<?php . . . ?>` tags.

6. What is the difference between `echo` and `print`?**Answer:**

`echo` can output multiple strings and is faster, while `print` returns a value and outputs only one string.

7. How do you declare variables in PHP?**Answer:**

Using the `$` symbol, e.g., `$name = "John";`

8. What are constants in PHP?**Answer:**

Constants are values that cannot be changed once defined, created using `define()`.

9. What data types are supported in PHP?**Answer:**

String, Integer, Float, Boolean, Array, Object, NULL, and Resource.

10. What is an array in PHP?**Answer:**

An array is a variable that can store multiple values in a single variable.

11. Types of arrays in PHP.**Answer:**

Indexed arrays, Associative arrays, and Multidimensional arrays.

12. What are operators in PHP?**Answer:**

Operators are symbols used to perform operations like arithmetic, comparison, and logical operations.

13. What are control structures in PHP?

Answer:

Control structures control the flow of execution, such as `if`, `else`, and `switch`.

14. What looping structures are available in PHP?

Answer:

`for`, `while`, `do-while`, and `foreach`.

15. What is a function in PHP?

Answer:

A function is a reusable block of code that performs a specific task.

16. How do you read user input data in PHP?

Answer:

Using superglobal arrays like `$_GET`, `$_POST`, and `$_REQUEST`.

17. How do you embed PHP within HTML?

Answer:

By placing PHP code inside HTML using `<?php ?>` tags.

18. How does PHP connect to a database?

Answer:

Using extensions like `mysqli` or `PDO`.

19. What is phpdbg?

Answer:

`phpdbg` is a built-in interactive debugger used for debugging PHP applications.

20. What are common PHP debugging techniques?

Answer:

Using `echo`, `print_r()`, `var_dump()`, error reporting, and `phpdbg`.

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1. What is a PHP framework?

Answer:

A PHP framework is a platform that provides a structured way to build web applications using reusable components and best practices.

2. Why should we use a PHP framework?

Answer:

It speeds up development, improves security, enforces MVC architecture, and makes applications easier to maintain.

3. What is MVC architecture?

Answer:

MVC stands for Model–View–Controller, a design pattern that separates application logic, user interface, and control logic.

4. What is the role of the Model in MVC?

Answer:

The Model handles data logic, database interaction, and business rules.

5. What is the role of the View in MVC?

Answer:

The View is responsible for displaying data to the user, usually using HTML templates.

6. What is the role of the Controller in MVC?

Answer:

The Controller handles user requests, processes input, interacts with the Model, and returns the View.

7. What is a naming convention?

Answer:

A naming convention is a set of rules for naming files, classes, methods, and variables to maintain consistency.

8. Give an example of naming convention in PHP frameworks.

Answer:

Controllers use PascalCase (e.g., `UserController`), models use singular nouns (e.g., `User`).

9. What is routing in PHP frameworks?

Answer:

Routing maps URLs to specific controller methods.

10. How does a PHP framework connect to a database?

Answer:

By configuring database credentials in a configuration file and using built-in database libraries or ORM.

11. What is ORM?

Answer:

ORM (Object Relational Mapping) allows database tables to be accessed using objects instead of SQL queries.

12. Name an ORM used in PHP frameworks.

Answer:

Eloquent (Laravel), Doctrine (Symfony), and CakePHP ORM.

13. What is database migration?

Answer:

Migration is a way to manage database schema changes using version-controlled files.

14. What is a query builder?

Answer:

A query builder provides an easy, secure way to write database queries programmatically without raw SQL.

15. How is database interaction handled in MVC?

Answer:

The Controller receives the request, the Model interacts with the database, and the result is passed to the View.

16. What is CRUD operation?

Answer:

CRUD stands for Create, Read, Update, and Delete operations on database records.

17. How do frameworks improve database security?

Answer:

By using prepared statements, ORM, and input validation to prevent SQL injection.

18. What is a configuration file?

Answer:

A configuration file stores application settings such as database credentials and environment variables.

19. What is dependency injection?

Answer:

Dependency injection is a technique where objects receive their dependencies from an external source rather than creating them internally.

20. What are the advantages of MVC-based PHP frameworks?

Answer:

Better code organization, easier maintenance, improved scalability, and cleaner separation of concerns.

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1. What are browser developer tools?

Answer:

Browser developer tools are built-in tools that help developers inspect, debug, and optimize web applications.

2. How do you open developer tools in a browser?

Answer:

By pressing **F12**, **Ctrl+Shift+I**, or right-clicking and selecting **Inspect**.

3. What is the DOM?

Answer:

DOM (Document Object Model) is a tree-like structure representing the HTML elements of a web page.

4. Which panel is used to view and modify the DOM?

Answer:

The **Elements** panel.

5. Can changes made to DOM in developer tools affect the actual source code?

Answer:

No, changes are temporary and only affect the current browser session.

6. How can you inspect and modify CSS using developer tools?

Answer:

By using the **Styles** section in the Elements panel.

7. What is the Console panel used for?

Answer:

It is used to display JavaScript errors, logs, warnings, and for executing JavaScript commands.

8. Name some common console methods.

Answer:

`console.log()`, `console.error()`, `console.warn()`, and `console.table()`.

9. How do you debug JavaScript using developer tools?

Answer:

By using the **Sources** panel and setting breakpoints.

10. What is a breakpoint?

Answer:

A breakpoint pauses JavaScript execution at a specific line to inspect variables and code flow.

11. What is step-through debugging?

Answer:

It allows executing code line by line to understand program behavior.

12. Which panel is used to view network activity?

Answer:

The **Network** panel.

13. What information can be seen in the Network panel?

Answer:

HTTP requests, responses, status codes, request headers, response headers, and load times.

14. How can you identify slow network requests?

Answer:

By checking request timing and waterfall charts in the Network panel.

15. What is the use of the Performance panel?

Answer:

It analyzes page load performance, rendering, scripting time, and CPU usage.

16. What are performance bottlenecks?

Answer:

Parts of the code that slow down page loading or responsiveness.

17. What is cache and how can it be disabled in developer tools?

Answer:

Cache stores resources locally; it can be disabled from the Network panel while DevTools is open.

18. What is live editing?

Answer:

Live editing allows real-time modification of HTML, CSS, or JavaScript for testing purposes.

19. What is the purpose of the Application panel?

Answer:

It is used to inspect cookies, local storage, session storage, and cache data.

20. Why are browser debugging tools important?

Answer:

They help identify errors, optimize performance, test changes quickly, and improve web application quality.